

1114I

Management Studies Tripos

**CORE MODULE
COURSEWORK**

MS5

Operations Management

Course Leader: Dr. Benn Lawson

Hand-in date:
26/04/21

1092

(1000)

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Introduction

Supply chains are the alignment of different organisations combining to deliver value to customers (Lambert, 1998). The pandemic has simultaneously disrupted global supply chains and induced demand variability, underscoring the need to understand how agile supply chain management can minimise costs during disruptions (Jack, 2020). This is the focus of my essay. I will also outline how organisations can leverage technology to better forecast demand fluctuations.

The Covid Challenge

Covid-19 has created many operational challenges as lockdowns and heightened border controls strain supply chains. Issues have been magnified for original equipment manufacturers (OEMs) lacking understanding of their supply chains. The confession of one executive that “I know it will hit us, I just have no idea how” (Campbell, 2020) underscores the lack of visibility many organisations have beyond immediate suppliers. This makes it difficult to mitigate against supply chain disruption. On the demand side, the pandemic has altered consumption trends. OEMs experiencing rising demand and supply chain disruptions risk losing customers to competitors as lead-times balloon. In its quarterly earnings statement, exercise equipment manufacturer Peloton revealed that ‘Covid-related factors continue to present challenges to returning our delivery times to pre-pandemic levels’ (Gurman, 2021).

Compensating for supply chain disruptions by modifying manufacturing processes can lead to quality issues. Some commentators suggest this was the source of Boeing’s woes with its 787 (Leggett, 2019). As such, my essay aims to instead explore how modifying supply chains can solve these issues.

Supply Chain Management

Supply chain management controls the sourcing and flow of materials by adopting a holistic perspective considering multiple functions and tiers of suppliers (Monczka, 1998). This alludes to Mentzer’s (2001) three degrees of supply chain complexity, with the most complex, the ‘*ultimate supply chain*’, including all organisations involved in product and information flows between the ultimate supplier and customer.

Slack and Brandon-Jones (2019) distinguish lean supply chains, emphasising efficiency, from agile supply chains, emphasising flexibility. They suggest ‘functional’ products require the former and ‘innovative’ products require the latter, due to the responsiveness necessary to meet variable demand. The pandemic has blurred this distinction, with functional products like exercise equipment having huge demand surges (Cision, 2020). Take Peloton, whose business model relies on ‘locking in’ customers to subscriptions. The window to do so is closing, as the reopening of gyms is dampening home workout trends (Cox, 2021). These market conditions highlight why Peloton should prioritise responsiveness over cost-efficiency, recognised in its decision to spend \$100m on air cargo to reduce lead-times. How can Peloton’s supply chain better match supply to demand, thereby avoiding the firm this expense?

Location, location, location

Agile supply chain management involves deploying inventory close to the customer to maintain supply during disruptions (Slack and Brandon-Jones, 2019). Presumably for this reason, Peloton has recently purchased Precor, a fitness equipment manufacturer based in North Carolina (McGee, 2021). However for some organisations, the storage costs and bullwhip effects associated with stockpiling (Lee, 1997) can outweigh the benefits. To develop agile supply chains that respond to demand and supply changes quickly, such organisations can instead promote information flows with suppliers and customers (Lee, 2004). This, and other theories, are now explored.

Diversify the Supply Base

Globalisation has induced many organisations to trade supply chain agility for cost-efficiencies (Campbell, 2020). A shift back towards agility would start with mapping ultimate supply chains to expose hidden vulnerabilities (Shih, 2020). Understanding the *ultimate* supply chain is critical; whilst reliance on specialised inputs from one supplier increases the OEM's vulnerability, so do suppliers that source their inputs from one organisation or location. Peloton could identify critical points in its supply chain using technologies such as radio frequency identification, which tag and track components throughout the chain. Peloton could then classify critical points based on predicted impacts on revenues (Shih, 2020). Machine learning can enhance prediction accuracies, for example by creating regression models. After conducting these analyses, Peloton should seek to develop multiple suppliers for key components, and create 'reserve suppliers' by auditing facilities. This strategy was deployed by Boll & Branch, allowing the firm to meet rising demand even as some suppliers shut down (Sorokanich, 2020).

Diversifying its supply base will give Peloton greater scope to maintain material inflows if a supplier is disrupted, or increase material inflows when demand rises. However, the challenge of mapping the ultimate supply chain for complex products is enormous (Keim, 2013).

Supplier Relationships: Communication is Key

So as well as leveraging the technologies described, Peloton could develop deeper relationships with its suppliers to further understand its ultimate supply chain. Applying Liker and Choi's (2004) supplier-partnering hierarchy, Peloton should seek to understand how its suppliers work, before helping develop their technical capabilities and sharing information intensively but selectively. This structured, clear communication will reduce the fulfilment perception gap and requirements perception gap outlined by Slack and Brandon-Jones (2019), allowing suppliers to better understand Peloton's needs, and therefore better assess their performance relative to those needs.

Forecasting: Creating Your Crystal Ball

These steps should give Peloton greater confidence in the ability of its suppliers to meet its needs. However, a supply chain that is agile the whole way through requires OEMs in turn being able to meet the needs of their customers. Developing forecasting capabilities will help here. Peloton could draw inspiration from Samsung, which is using AI-based forecasting and trend analysis, supplied with Covid-specific information, to predict regional demand more accurately (White, 2021). When coupled with consistent, controlled communication with suppliers, enhanced forecasting capabilities can also help reduce the bullwhip effect (Slack and Brandon-Jones, 2019). Developing systems to control the bullwhip effect should rise up Peloton's to-do list once it has addressed its more existential supply chain issues.

Concluding Remarks

This essay has explored why and how Peloton should increase the agility of its supply chain to meet rising demand. Both quantitative and qualitative techniques should be employed. Leveraging data analytics can enhance Peloton's demand forecasting and understanding of its risk exposure throughout its ultimate supply chain. This information allows organisations to model the impact of disruptions, thus guiding the allocation of resources to develop alternative suppliers and distribution channels. Qualitatively, deeper supplier relationships will help map ultimate supply chains, and can increase the quality of material inflows.

Lee, author of the seminal 'Triple-A Supply Chain' (2004), suggests most organisations already have the infrastructure to achieve supply chain agility – they simply need to give up their 'efficiency mindset'. Whilst this may not benefit all organisations, moving from supply chain efficiency to agility could certainly be a solution to Peloton's problems because responsiveness complements the organisation's business model and current market dynamics.

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Introduction

During the pandemic, organisations have modified operations in response to government regulations and variability in customer demand and material inflows. Aside from these fluctuations, the variability of labour due to isolation rules presents a more unconventional challenge.

The pandemic could be an impetus to embrace new process technologies that can increase process efficiency and resilience, whilst delivering cost-savings. Slack and Brandon-Jones (2019) categorise process technologies based on five primary capabilities: thinking, seeing, communicating, materials processing and transportation. This essay will introduce several technologies with such capabilities, exploring how they could modify processes to solve Covid-related operational challenges. Following this, I explore the factors managers should consider when evaluating new technologies.

Additive Manufacturing

Additive manufacturing layers materials to make objects from 3-d model data (Bikas, 2015). It is significantly less wasteful than subtractive manufacturing, which uses material removal processes (Ford, 2016). This reduces the amount of inventory needing to be held and cuts lead times (Gaget, 2018), making additive manufacturing conducive to lean production systems.

Linking to this, General Electric has used additive manufacturing to slash lead times for its Advanced Turboprop engine by reducing 855 subtractive components to 12 additively manufactured components (Zaleski, 2015). The implications of this are profound; aside from increasing production speed without trading off quality, it gives GE scope to improve its supply chain management. Reducing the number of components used can in turn reduce the number of suppliers GE relies upon, thereby making it easier for the organisation to understand its ultimate supply chain (Mentzer, 2001) and develop contingency plans accordingly. The absence of this planning has been a key contributor to Covid-related supply chain troubles (Shih, 2020).

That said, in the technology's current state, only polymers, ceramics and metals can be additively manufactured (Colorado, 2020). What process technologies could be utilised by a wider range of organisations?

Process Automation

As well as changing automated processes, technology can automate labour-intensive processes. Automation can increase the speed and accuracy of processes, translating into higher output quality, improved safety, and a more efficient use of resources (Jamsa-Jounela, 2007). This is common knowledge, but technological maturation, coupled with Covid-specific operational challenges, might accelerate automation trends.

The automation of inventory tracking, logistics and equipment monitoring is increasingly viable due to the maturation of flying drone technologies (Jackson, 2017). Aside from significant improvements in speed and error rates (Geissbauer), this process technology can radically reduce the frequency of human contacts in warehouses and manufacturing facilities. This is key in the current context, as it can enable regulatory compliance. This can save organisations time, by removing the need for one-way employee traffic systems, and money, by avoiding the need to supply PPE equipment.

Digital Twins

Another emerging process technology is the digital twin, whereby virtual copies of products or processes are constructed in simulations to assess their performance (Lawson, 2020). Developing prototypes in digital twins saves time and money by reducing the number of physical prototypes (Siemens); the feedback loop of performance data in the simulation means the first physical prototype is many iterations ahead of what might otherwise be produced.

Focusing on Covid, digital twins can reduce the need for specific processes that can be performed in the simulation. In this sense the digital twin can be a contingency plan, substituting for labour inputs if key employees are isolating, or new regulations increase the cost of performing a process. To illustrate, GM recently used digital twins to cut the development time of its Hummer EV from five to two years, performing virtual test drives, safety checks and predictive maintenance (Valdes-Dapena, 2020).

The scalability and connectivity of digital twins is exciting; Geissbauer highlights the potential to simulate whole factory lines, which leads me to the ability of digital twins to integrate other emerging technologies. For example, simulation conditions can be informed by data collected via machine learning, either during production or during consumption, using sensory technology. Tying scalability and connectivity together, organisations could therefore construct digital twins of production lines, which are then used to accurately assess the impacts of new process technologies *before* capital expenditure.

The Scope for Digitalisation

Process digitalisation is not a quick fix. The benefits of the technologies outlined, including efficiency gains and reduced labour dependence, must be evaluated on a case-by-case basis.

Managers must factor in the role of customers to their assessment of a new process technology. Whether its introduction will add value for customers will depend upon customers' ability to use the technology, and the order-winners and qualifiers of the specific process. To illustrate, the increased speed and social distancing allowed by digitalised ordering in MacDonald's (Kowitt, 2020) adds value for customers. However, automating the post-operative assessments carried out by nurses, to increase throughput rate thus capacity, would reduce customer value; the nuanced process cannot be replicated through questionnaires and, crucially, patients value attention and speed with opposite valence to fast-food customers.

Additionally, automation can make employees feel marginalised, sapping morale (Geissbauer). This could increase the costs of adopting new technologies, for example if labour productivity in other processes drops. Further, where these sentiments become ingrained, they can translate into cultural resistance to technological change. Managers must be careful not to have their evaluations of emerging technology unduly biased by these factors. These concerns link to Marque's (2020) finding that when implementing new process technology, mature companies face additional challenges of ingrained thinking and self-imposed bureaucracy. He suggests this can be overcome through a zero-based process design, which demands fresh justification to support each process.

Concluding Remarks

New process technologies can solve Covid-related challenges, as explored in this essay. However when evaluating a new technology, managers must minimise the risk of organisational bias with zero-based process designs, whilst remaining attentive to employee sentiments and being bound by customer needs.

They must also consider the current needs of the process and the technology's capabilities, in order to assess the 'fit' between the technology and the process' position on the product-process matrix (Hayes and Wheelwright, 1979). Revisiting the nursing example, this is not a matter of simply dismissing the use of new technology for high-variety, low-volume processes – it is a matter of finding the *right* technology to augment these processes. Slack and Brandon-Jones (2019) believe this will typically be technology that has a low degree of automation and scalability, and does not have to be coupled with other technologies.

The pandemic might colour these evaluations in favour of new technologies because of the additional costs associated with labour-intensive processes. Nevertheless, paying attention to the factors discussed should allow managers to implement the right process technologies, in the right manner, to achieve the right results.

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